

ELITE IGCSE MATHEMATICS  
EXPERIENCE NOTES

# 3D Pythagoras & Trigonometry

*Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.*

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3D PYTHAGORAS & TRIGONOMETRY · CH4 · L12

ENG. ESLAM AHMED



# The Map

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This lesson collapses 3D Pythagoras & Trigonometry into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

**01** Draw the hidden right triangle

TRIGGER *"draw", "plot", "construct", a diagram/grid*

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**02** Find the base diagonal first

TRIGGER *"work out", "find", a missing value*

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**03** Use 3D trig after Pythagoras

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

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**04** Give the final angle/length

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

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**05** Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

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**06** Find the midpoint

TRIGGER *"work out", "find", a missing value*

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BANK EVIDENCE    29 website questions, 29 checked solutions, 3 local PDFs read.

# 01 Draw the hidden right triangle

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A 3D Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\angle ABM = \frac{140^\circ}{2} = 70^\circ$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

**Source pattern:** Jan 2021 P2HR Q22

**Use trigonometry**

This is a reasoning step: write one clear sentence, then continue the calculation.

**The line from to the**

$$\angle ABM = \frac{140^\circ}{2} = 70^\circ$$

**Finish:** 74.7°

FORMULA BANK

Make a 2D right triangle inside the 3D shape, then use Pythagoras/SOH CAH TOA on that triangle.

**BOTTOM LINE**

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

**SAME IDEA** Use the same first line from Strategy 01.

## QUESTION

A cuboid is 8 cm by 6 cm by 12 cm. Find the length of the space diagonal, correct to 3 significant figures.

## YOUR SOLUTION

## 02 Find the base diagonal first

TRIGGER *"work out", "find", a missing value*

BECOMES A 3D Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\angle ABM = \frac{140^\circ}{2} = 70^\circ$$

SIMPLEST STRATEGY

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**BOTTOM LINE**

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.



## 03 Use 3D trig after Pythagoras

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A 3D Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\sqrt{9^2 + 5^2} = \sqrt{106}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

**Source pattern:** Jan 2020 P1H Q19

**The projection of onto the**

$$\sqrt{9^2 + 5^2} = \sqrt{106}$$

**Use trigonometry**

$$\tan \theta = \frac{6}{\sqrt{106}}$$

**Finish:** 30.2°.

FORMULA BANK

Make a 2D right triangle inside the 3D shape, then use Pythagoras/SOH CAH TOA on that triangle.

**BOTTOM LINE**

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

**SAME IDEA** Use the same first line from Strategy 03.

## QUESTION

A cuboid is 8 cm by 6 cm by 12 cm. Find the length of the space diagonal, correct to 3 significant figures.

## YOUR SOLUTION

## 04 Give the final angle/length

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A 3D Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$AB = 15, \quad BC = 4, \quad CF = 9$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

**Source pattern:** Jun 2023 P2HR Q22

**Find the gradient**

$$AB = 15, \quad BC = 4, \quad CF = 9$$

**Use trigonometry**

$$PR = 42$$

**Finish:**  $BE = 17.9 \text{ cm}$ ,  $\angle UMR = 49.1^\circ$

FORMULA BANK

Make a 2D right triangle inside the 3D shape, then use Pythagoras/SOH CAH TOA on that triangle.

**BOTTOM LINE**

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

**SAME IDEA** Use the same first line from Strategy 04.

## QUESTION

A cuboid is 8 cm by 6 cm by 12 cm. Find the length of the space diagonal, correct to 3 significant figures.

## YOUR SOLUTION

## 05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A 3D Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\sqrt{9^2 + 5^2} = \sqrt{106}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

**Source pattern:** Jan 2020 P1H Q19

**The projection of onto the**

$$\sqrt{9^2 + 5^2} = \sqrt{106}$$

**Use trigonometry**

$$\tan \theta = \frac{6}{\sqrt{106}}$$

**Finish:** 30.2°.

FORMULA BANK

Make a 2D right triangle inside the 3D shape, then use Pythagoras/SOH CAH TOA on that triangle.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.



## 06 Find the midpoint

TRIGGER *"work out", "find", a missing value*

BECOMES A 3D Pythagoras & Trigonometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$AE = x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

**Source pattern:** Jan 2022 P1H Q21

**Let**

$$AE = x$$

**Use trigonometry**

$$AB = BD = DE = AE = x$$

**Finish:**  $T = \frac{\sqrt{68} + \sqrt{51}}{17}$

FORMULA BANK

Make a 2D right triangle inside the 3D shape, then use Pythagoras/SOH CAH TOA on that triangle.

**BOTTOM LINE**

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.



# Quick Reference

TRIGGER → FIRST MOVE

| WHEN YOU SEE                                              | WRITE THIS FIRST               |
|-----------------------------------------------------------|--------------------------------|
| "draw", "plot",<br>"construct", a dia-<br>gram/grid       | Draw the hidden right triangle |
| "work out", "find", a<br>missing value                    | Find the base diagonal first   |
| "work out", "show",<br>"hence", a familiar<br>exam phrase | Use 3D trig after Pythagoras   |
| "work out", "show",<br>"hence", a familiar<br>exam phrase | Give the final angle/length    |
| "work out", "show",<br>"hence", a familiar<br>exam phrase | Use trigonometry               |
| "work out", "find", a<br>missing value                    | Find the midpoint              |

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.